

Trainings ▾

General Information

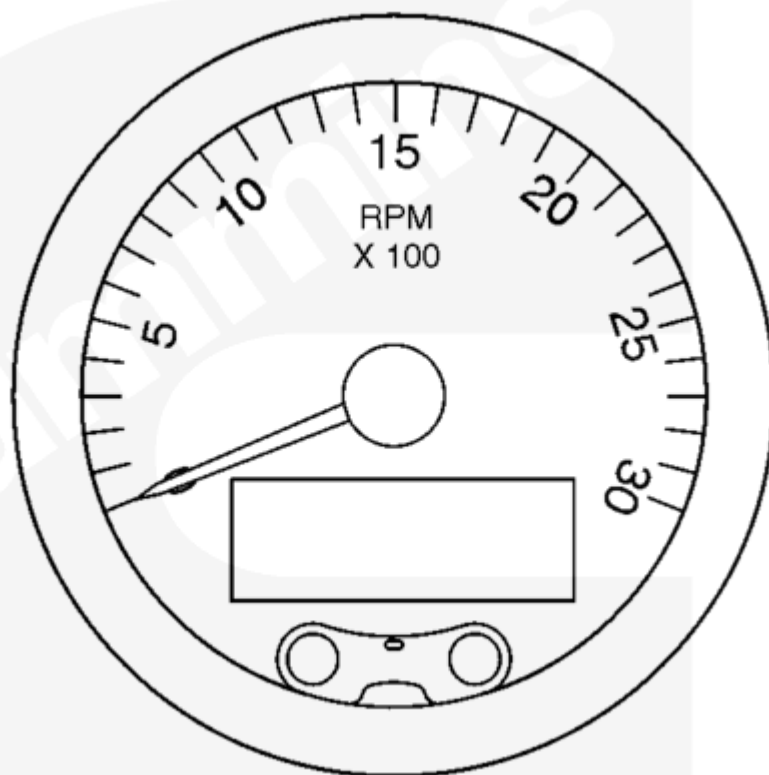
The gauges have been developed for applications where the SAE J1939 datalink communication bus is available.

The gauge components of the system are based on stepper motor technology, integrated with a microcontroller that provides the necessary electrical control. All gauges are constructed using the standard size of 52 mm [2.05 in] diameter. The instrument enclosure can have the connector molded in as an integral part. The gauge sets have one master gauge and up to six slave gauges.

The master gauge includes all communications, power supply, memory, display, and electronics to control the system.

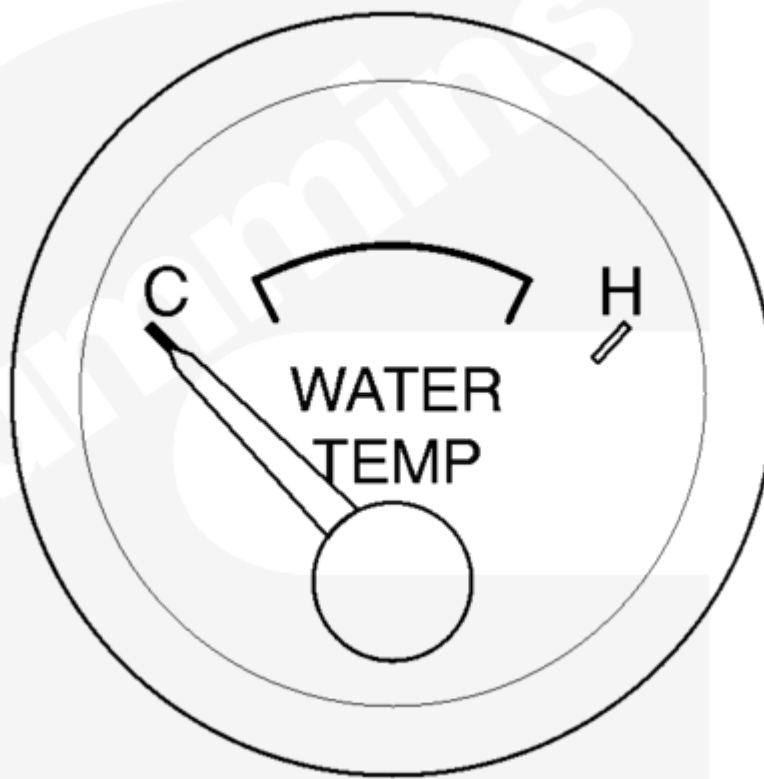
The slave gauges, including the warning bank modules, will receive power and communications from the master device.

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17800082

RPM Gauge (Master)



17800083

Coolant Temperature Gauge (Slave)

Lighting of the dial face is provided by light emitting diodes. The dial is a through illumination. The backlighting of the gauge is controlled by the gauge's microcontroller via a pulse width modulated output. Standard backlight color is red.

The pointer length is 44 mm [1.73 in]. The pointer is illuminated when the gauge backlight is on. Standard pointer color is red. The pointer can perform a 230 degree sweep. The pointer hub is black.

In each gauge a microprocessor controlled stepper motor will position the pointer to within an angular degree of the position determined by the data received from the SAE J1939 datalink over the entire operating temperature and voltage range.

The message center is a full graphic display. The specifications are described in the table below:

Gauge Display Specifications	
Description	Specification
Viewable area	51.6 mm [2.15 in] x 19.05 mm [0.75 in]
Pixel array	122 x 32 pixels
Polarizer mode	Transflective positive
Backlighting	Custom
Operating temperature	-20°C [-4°F] to 70°C [158°F]
Storage temperature	-40°C [-40°F] to 85°C [185°F]
Backlight brightness	Pulse width modulated controlled by gauge's microprocessor

The user can interact with the gauge set through the mode and trip buttons on the front of the master gauge.

There is an audible alarm in back of the master gauge. The tones are assigned to warning conditions received from the SAE J1939 datalink. Each slave gauge has a lamp that illuminates when an alarm occurs.

There are two communications channels for data transfer between the vessel and master gauge. One channel for SAEJ1708/J1587 datalink and second channel for SAE J1939 datalink. A selectable termination resistor is available for the SAE J1939 datalink.

A local interconnect network communication channel is used for communication between the master gauge and the slave gauges.

The main power input for the master gauge is 9 to 32 VDC for normal operation. Voltage greater than 9 VDC will cause the microcontroller to shut down because of low voltage.

The main power input for the slave gauge is 7.5 VDC and supplied by the master gauge. This supply is protected against EMC, over voltage, reverse polarity, and short circuit.

The following table are for connector assignments:

Connector to Slave Devices		
Terminal	Designation	Description
1	(+) Power	7.5 VDC power output
2	GND	Device ground
3	LIN	Local interconnect network for slave functions
4	Open	Not used

Power and Communication Connector		
Terminal	Designation	Description
1	(+) Battery	Unswitched positive battery input
2	Ground	Chassis ground
3	(+) CAN	SAE J1939 datalink supply
4	(-) CAN	SAE J1939 datalink return
5	(+) J1708	SAE J1708 datalink supply
6	(-) J1708	SAE J1708 datalink return
7	Reserved	Reserved for ISO 9141 diagnostic communication port.
8	CAN Res	SAE J1939 datalink termination resistor.

Master Gauge Input and Output Connector			
Terminal	Designation	Type	Description
1	AN3	Voltage or resistance	Analog input
2	AN1	Voltage or resistance	Analog input
3	AN2	Voltage or resistance	Analog input
4	AN0	Voltage or resistance	Analog input with wake-up capability
5	AN4	Voltage or resistance	Analog input
6	AN5	Voltage or resistance	Analog input
7	AN6	Voltage or resistance	Analog input
8	AN7	Voltage or resistance	Analog input
9	BIN_OUT1	Low side 1 amp	Binary output
10	BIN_OUT2	Low side 1 amp	Binary output
11	IGNITION	High side	Binary input for ignition switch
12	BIN0	Pull-up or pull-down	Binary input with wake-up capability
13	BIN1	Pull-up or pull-down	Binary input with wake-up capability
14	BIN2	Pull-up or pull-down	Binary input
15	BIN3	Pull-up or pull-down	Binary input
16	BIN4	Pull-up or pull-down	Binary input